

Kernel Boot Begins

Limine Bootloader provides memory map
(regions marked as USABLE, RESERVED...)

Identify memory ranges marked as USABLE
in the memory map

Calculate total number of frames (4 KiB size)

Allocate memory for Bitmap

- 1 bit = 1 frame
- Bitmap size = total_frames / 8

Initialize Bitmap

- Set bit = 0 → frame is free
- Set bit = 1 → frame is used

Mark reserved regions as used

- Kernel ELF sections (text, data)
 - Framebuffer memory
- Page tables & stack memory

allocateFrame()

- Find first 0 bit (free frame)
- Set bit to 1 (mark as used)
- Return frame index * PAGE_SIZE

freeFrame(physical_address)

- Convert physical addr → frame index
- Set bit to 0 (mark as free)

Return usable physical address (4 KiB
aligned)